Mediation in Regression Discontinuity: A Replication of Chen (2021)

November 2026

Abstract

We did not replicate three of the four simulation headline findings of Chen (2021), who reported that mediation analysis in a sharp regression discontinuity design gives biased natural direct effects, percentile bootstrap confidence intervals that cover the true value far less often than the nominal level, and near-zero power for the mediated interaction: in all 3,840 condition-cutoff cells, 0.00% of our direct-effect results had |bias| > .10 against 85.42% in our average of her six direct-effect Table 4.8 rows; coverage fell 0.04 rather than 43.62 percentage points short of 95%; the truth lay above the interval 0.33 rather than 47.07 points more often than below it; and mediated-interaction power at n = 500 for the largest interaction (a = .59, h = .34) was 0.836 where she reported near zero. We could not establish why her results differ, as the step comparing her estimates with population values is unprinted and her result files are unavailable; her Head Start comparison is not established (no data). Her abstract puts Type I error and power within the robustness criteria, against her near-zero mediated-interaction power; 29 of the 46 statements we tested from her text agree. Her tentative minimum n of 5,000 was partly replicated: 86.3% of zero-effect results have Type I error in range and 89.2% of nonzero results reach .80 power, but effects with both paths zero stay conservative.

Claim / estimand	Published	Ours	Published effect	Our effect	Effect magnitude	Direction	Verdict
H1, E1. Percentage of PNDE and TNDE condition-cutoff results with |bias| > .10, averaged over the six effect-cutoff rows of Table 4.8	PNDE and TNDE within [-.10, .10] in only 12.50-18.75% of conditions, with negative bias	PNDE and TNDE within [-.05, .05] in 100% of conditions; largest |bias| 0.016	85.42	0.00	smaller	zero vs nonzero	no match
H2, E2. 95 minus mean coverage (%) over the 15 effect-cutoff rows of Table 4.9	Coverage 24.10-83.99% by effect and cutoff; true value usually above the CI	Coverage 94.6-95.6% by effect and cutoff; misses on both sides at similar rates	43.62	0.04	smaller	same direction	no match
H2, E3. Mean over the 15 Table 4.9 rows of (% of replications with the population value above the CI) minus (% with it below the CI)	Coverage 24.10-83.99% by effect and cutoff; true value usually above the CI	Coverage 94.6-95.6% by effect and cutoff; misses on both sides at similar rates	47.07	0.33	smaller	same direction	no match
H3, E4. MINT rejection rate at n = 500, a = .59, h = .34, mean over b, c' and cutoffs. The source states a verbal result; a scalar contrast exists but no defensible margin.	MINT power near zero at every sample size; never reaches .80	MINT power 0.836 at n = 500 for ah = .20; at least .97 for ah >= .13 from n = 1,000	not established	0.836	not established	not established	no match
H4, E5. qualitative sample-size recommendation over the grid. A qualitative design recommendation over the whole grid; no single signed contrast represents it, and any pooled share would depend on the mix of effect sizes in the grid.	Tentative minimum n of 5,000 for accurate Type I error and power above .80	At n = 5,000, 86.3% of zero-effect results have Type I in range and 89.2% of nonzero results reach .80 power, similar at each cutoff; both-zero products stay conservative	not established	not established	not established	not established	partial

Table 1. Headline claims of Chen (2021). PNIE, TNIE, PNDE, TNDE: pure and total natural indirect and direct effects; MINT: mediated interaction; condition-cutoff cell: one of 1,280 conditions at one of three cutoffs; both-zero products: effects with both paths zero (Type I error near .002). Published: Chen's Tables 4.8, 4.9 and text (printed pp. 88, 121, 124, 147, 148, 150); E1–E3 published effects are our averages of her rows; effects are signed. Tolerances and units: E1 tolerance 0.005 percentage points; E2 tolerance 0.005 percentage points; E3 tolerance 0.01 percentage points; E4 tolerance not established, units rejection rate; E5 tolerance not applicable, units not applicable. Chen reports no Monte Carlo errors; ours are about 0.7 percentage points per cell for coverage and rejection (E2–E4) and at most 0.006 for bias (E1). Descriptive point estimates; not equivalence. This applies to the magnitude labels of rows E1–E4; E5 is qualitative.

1 Introduction

We did not replicate three of the four simulation headline findings of Chen (2021), who reported that mediation analysis in a sharp regression discontinuity design (RDD) gives biased natural direct effects, percentile bootstrap confidence intervals (CIs) that cover the true value far less often than the nominal 95%, and near-zero power for the mediated interaction: in all 3,840 of 3,840 condition-cutoff cells (each of 1,280 simulated conditions analyzed at each of three cutoffs), 0.00% of our direct-effect results had $|\text{bias}| > .10$ against 85.42% in our average of the six direct-effect rows of her Table 4.8, mean coverage fell 0.04 rather than 43.62 percentage points short of 95%, the population value lay above the CI 0.33 rather than 47.07 percentage points more often than below it (Table 4.9), and mediated-interaction power was 0.836 at $n = 500$ for the largest mediated interaction in the design ($a = .59$, $h = .34$; 0.011–0.719 for the smaller ones) where she reported that "All power for MINT is near zero" (Chen, 2021, printed p. 121). We could not establish why her printed bias, coverage and mediated-interaction results differ from ours, because the step that compared her estimates with the population (true) values is not printed and her result files are not available; her second research question, which compares RDD and randomized estimates in the Head Start Impact Study (HSIS), is not established because the restricted-use data were unavailable to us. No headline effect reversed its sign: our coverage shortfall and imbalance have the same sign as hers but are about 140 times (imbalance) and over 1,000 times (shortfall) smaller, and our share of direct-effect results outside $[-.10, .10]$ is zero where hers is 85.42%. Her own abstract concluded that "Type I error rate and power for all mediation effects were within or reached the robustness criteria, but bias and coverage were more often outside the robustness range" (Chen, 2021, abstract, PDF p. 3); 29 of the 46 Type I error and power statements we tested from her text agree with our run (31 when every threshold is moved .01 in her favour; Section 5), including all of her PNIE and PNDE power statements, and 39 would hold if, in our results, the treatment-by-mediator interaction h were set to 0 (a reading of our results, not a test of her runs); her bias, coverage and mediated-interaction results do not agree, and her abstract's power statement conflicts with her Chapter 4–5 finding of near-zero mediated-interaction power. Her tentative recommendation of a minimum sample size of 5,000 (Chen, 2021, printed p. 150) was partly replicated. The comparison tolerance in Table 1 is half a unit in Chen's last printed decimal, carried through the averages that form each contrast (0.005, 0.005 and 0.01 percentage points); it is a rounding bound, not a sampling band, and the magnitude labels are descriptive point-estimate comparisons, not equivalence tests.

Chen's dissertation asks three research questions (Chen, 2021, printed p. 7): what assumptions mediation requires in an RDD (RQ1, answered by the literature synthesis in her Table 4.1); how mediation in an RDD compares with mediation in a randomized controlled trial (RCT), using the HSIS (RQ2); and how mediation in an RDD behaves across sample size, path effect sizes and cutoff in a Monte Carlo simulation (RQ3). Her headline conclusions come from RQ3 and are restated in her Chapter 5: biased direct effects (printed p. 147), invalid percentile bootstrap CIs with the truth "systematically" above them (printed p. 150), near-zero power for the mediated interaction (printed p. 121; "extremely low" on p. 148), and the minimum sample size of 5,000 (printed p. 150). Section 4 answers each in turn. Section 5 covers her Type I error and power statements, RQ1 and RQ2. We label her four headline claims H1–H4 and her eight supporting claims S1–S8; Appendix C lists them all. Sections 7–12 record our assumptions, the ambiguities we resolved, what remains open, availability and next steps.

2 What we did ourselves, and what we took from others

The question here is which parts of this note are Chen's and which are ours. Chen's contribution is the model, the design and the published tables: the single-mediator model with a treatment-by-mediator interaction, the natural-effect estimators (her equations 3.5–3.11, printed pp. 53–54), the factorial design

of her Table 3.1 (printed p. 55), the R code of her Appendices A–B (printed pp. 175–189), and every published number we quote, each labeled with its table and printed page. The natural direct and indirect effects and the mediated interaction follow the potential-outcomes definitions she adopts from VanderWeele (2015) and Valeri and VanderWeele (2013) (her printed pp. 19 and 29). The robustness bands come from her sources: $[.025, .075]$ for Type I error and $[.925, .975]$ for coverage (Bradley, 1978), and $[-.05, .05]$ for bias (Hoogland and Boomsma, 1998), with her looser $[-.10, .10]$ (printed pp. 57–58).

Our contribution is an independent re-implementation of her simulation, its full run, and the comparison. Every number labeled "ours" is our simulation or calculation from the files in Appendix A; numbers we derived from Chen's printed values (the headline contrasts E1–E3, and the recomputed differences of her Tables 4.6–4.7) are labeled as our calculation from her printed figures. We supplied the assumptions listed in Section 7. We wrote no proofs; this replication is computational.

3 Methods

3.1 Data generation and estimation

The simulation follows Chen's Appendix A (printed pp. 175–178) and Appendix B (printed pp. 179–189), re-implemented in `chen_sim.py`. In each replication, $x \sim N(0,1)$ for n units; for each of three cutoffs at the 25th, 50th and 75th sample percentiles of x, $t = 1$ if x is at or above the cutoff, $r = x - \text{cutoff}$, $m = at + e_m$ with $e_m \sim N(0, 1-a^2)$, and $y = c't + bm + h\,tm + e_y$ with $e_y \sim N(0, 1-b^2)$; fresh residuals are drawn for each cutoff, as in Appendix A. Ordinary least squares fits $m \sim 1+t+r$ and $y \sim 1+t+r+m+tm$, and the effects are PNIE $= ab$, TNIE $= ab + ah$, PNDE $= c' + h\,i_{0M}$, TNDE $= c' + h\,i_{0M} + ah$ and MINT $= ah$, where i_{0M} is the intercept of the m model (Chen's equations 3.7–3.11). Here PNIE and TNIE are the pure and total natural indirect effects, PNDE and TNDE the pure and total natural direct effects, and MINT the mediated interaction.

Each replication forms 95% percentile bootstrap CIs (Efron and Tibshirani, 1993) from 500 resamples of the rows; as in Appendix B, the same resample indices serve all three cutoffs. The design crosses $n \in \{500, 1000, 2500, 5000, 10000\}$, $a, b, c' \in \{0, .14, .39, .59\}$ and $h \in \{0, .05, .15, .34\}$: 1,280 data conditions, each analyzed at three cutoffs, give 3,840 condition-cutoff cells, and each was run at Chen's 1,000 replications (Chen, 2021, printed p. 57).

3.2 Outcome measures

Following Chen (printed pp. 57–58), bias is the mean of 1,000 estimates minus the population value; a rejection is a CI that excludes zero, read as Type I error when the population effect is zero and as power otherwise; coverage is the share of CIs that contain the population value; and imbalance splits the misses into CIs lying wholly below the population value ("truth above CI") and wholly above it. `aggregate.py` collapses the cell results over sample size and effect sizes by effect and cutoff index, as in Chen's Tables 4.8 and 4.9.

3.3 Software and validation

R could not be installed on either of the two computers used (`fallback.json`; Section 8.3), so the simulation is written in Python with NumPy (Harris et al., 2020): Python 3.13.18 with NumPy 2.5.3 on a Linux x86_64 computer (401 conditions), and Python 3.10.12 with NumPy 2.2.6 on a Linux aarch64 computer (879 conditions), as recorded in `run-record.md`. Chen used R 4.0.3 (R Core Team, 2021). Python and NumPy were chosen because her estimator is plain OLS with percentile bootstrap CIs,

which needs no structural-equation package. Each condition has its own seed, `SeedSequence(20211201, spawn_key=(k,))`, so splitting the run across the two computers changes no result; two conditions computed on one and rerun on the other differed by at most 1.1×10^{-16} in any summary statistic. The run took 13.7 core-hours.

`validate_estimator.py` checks the estimator independently: its point estimates agree with scikit-learn 1.9.1 (Pedregosa et al., 2011) linear-regression fits to 1.1×10^{-14} over 20 random datasets, its weighted-count bootstrap agrees with explicit row resampling refitted by least squares over 100 resamples, and at $n = 200{,}000$ its estimates are within sampling error of the population values. The validation ran with scikit-learn 1.9.1 on the x86_64 computer (`run-record.md`).

3.4 Comparability of the two simulations

Table 2 records, for each side, the elements that determine whether a summary is comparable. The two sides share the model, the estimators, the bootstrap rule and counts, and the design; they differ in engine, random streams, and in that Chen's population values and aggregation code are not printed.

Table 2: Comparability record for Chen's simulation and ours. Chen's entries are taken from her Chapter 3 and Appendices A–B; ours from `chen_sim.py` and `aggregate.py`.

Element	Chen (2021)	Ours
Engine	R 4.0.3, `lm()`	Python and NumPy, normal equations
Estimator	OLS, $m \sim t + r$, $y \sim t + r + m + tm$	same
Missing data	none in the simulation	none
n, replications, resamples	500–10,000; 1,000; 500	same
CI rule	95% percentile bootstrap	same (R quantile type 7)
Random streams	R seeds 18374788 and 95409	NumPy, seed 20211201 per condition
Population values	not printed	derived (Section 7.1)
Aggregation	"collapsed across sample and effect size"	mean over 1,280 conditions per effect and cutoff

4 Results for the headline claims

4.1 H1: are the direct effects biased?

No: in our run every direct-effect condition mean lies within $\pm.05$ of its population value (largest |bias| .016), with no bias beyond Monte Carlo error, whereas Chen reported that PNDE and TNDE fall inside $[-.10, .10]$ in only 12.50–18.75% of conditions and that "When bias was present, the measure of bias was a negative value" (printed p. 147). Table 3 sets her Table 4.8 beside ours. All 1,280 conditions lie inside $[-.05, .05]$ for every effect and cutoff; the largest absolute bias over all 19,200 condition-cutoff-effect results is 0.016 (0.01551 before rounding; `summary.json`), against a Monte Carlo standard error of the mean of 1,000 estimates of at most 0.006 in any cell (the largest standard deviation over $\sqrt{1000}$ in `cells.csv`). The headline contrast E1, the share of direct-effect results with |bias| $> .10$ averaged over the six direct-effect rows, is 85.42 percentage points in our average of her printed rows and 0.00 in ours.

Chen suggested that the bias "may be an indicator" of global estimation and an incorrect functional form, calling the latter a "plausible explanation" (printed p. 147). In her data-generating model the assignment variable enters neither M nor Y (Appendix A, printed pp. 175–177), so the linear models she fits are

correctly specified and OLS has no source of bias; our results, with no bias beyond Monte Carlo error, are what that model implies. Her printed estimation code (Appendix B) implements the same estimator as ours; where the difference arises is unresolved (Section 10).

Table 3: Percentage of conditions with bias inside $[-.10, .10]$ and $[-.05, .05]$, by effect and cutoff index. "Chen" columns are copied from her Table 4.8 (printed p. 88); "ours" columns and the largest absolute bias are our simulation (`table48_ours.csv`), each over 1,280 conditions. Cutoff index follows the code (Section 8.1).

		Within $[-.10, .10]$ (%)		Within $[-.05, .05]$ (%)		Largest
Effect	Cutoff	Chen	Ours	Chen	Ours	|bias|, ours
PNIE	1	62.50	100.00	50.00	100.00	0.0081
PNIE	2	75.00	100.00	50.00	100.00	0.0070
PNIE	3	81.25	100.00	75.00	100.00	0.0079
TNIE	1	51.88	100.00	41.88	100.00	0.0118
TNIE	2	60.00	100.00	42.19	100.00	0.0093
TNIE	3	70.94	100.00	54.69	100.00	0.0113
PNDE	1	12.50	100.00	6.25	100.00	0.0143
PNDE	2	12.50	100.00	8.75	100.00	0.0123
PNDE	3	18.75	100.00	10.00	100.00	0.0155
TNDE	1	12.50	100.00	6.88	100.00	0.0144
TNDE	2	12.50	100.00	7.50	100.00	0.0117
TNDE	3	18.75	100.00	7.19	100.00	0.0146
MINT	1	87.50	100.00	75.00	100.00	0.0044
MINT	2	87.50	100.00	75.00	100.00	0.0064
MINT	3	87.50	100.00	75.00	100.00	0.0065

4.2 H2: do percentile bootstrap CIs give valid inference?

Yes in our run, contrary to Chen's conclusion that "the current method of bootstrapping and constructing CIs does not produce valid inferences for estimates" (printed p. 151), which she calls the chief conclusion of her study. Table 4 sets her Table 4.9 beside ours. Our mean coverage is 94.6–95.6% in every effect-cutoff row, against her 24.10–83.99%; overall, mean cell coverage is 94.96%, the lowest cell coverage is .916, and 92.9% of the 19,200 cell coverages lie inside Bradley's $[.925, .975]$. Misses fall on both sides: the truth lies above the CI in 2.5–2.9% of replications and below it in 1.8–2.7%. The resulting imbalance (E3) is 0.33 percentage points in our run against Chen's 47.07, and the coverage shortfall (E2) is 0.04 against her 43.62 (both her values are our calculation from her printed table). Our small positive imbalance has her sign but is two orders of magnitude smaller than hers; we did not investigate its cause. E3 in Table 1 is the "Above CI" minus "Below CI" columns of Table 4, averaged over its rows.

The gaps lie far outside Monte Carlo error. The smallest coverage gap between a printed row and ours is 11.14 percentage points (PNIE, cutoff 3: 83.99 against 95.13), about 16 times the binomial standard error of a single cell's coverage (about .007 with 1,000 replications); each row mean averages 1,280 cells and has a smaller error still. No multiplicity adjustment, such as Holm's over the 15 rows, could bring any row inside a two-standard-error band.

Table 4: Coverage and imbalance (%) by effect and cutoff index. "Chen" columns are copied from her Table 4.9 (printed p. 124); "ours" columns and the lowest cell coverage are our simulation (`table49_ours.csv`), each a mean over 1,280 conditions. "Below CI" means the truth lies below the interval.

		Below CI		Coverage		Above CI		Lowest cell
Effect	Cutoff	Chen	Ours	Chen	Ours	Chen	Ours	coverage, ours
PNIE	1	1.21	2.11	59.07	95.22	39.72	2.67	0.917
PNIE	2	1.59	2.22	70.94	95.15	27.47	2.63	0.919
PNIE	3	1.93	2.24	83.99	95.13	14.08	2.63	0.924
TNIE	1	0.72	2.40	41.97	94.77	57.31	2.83	0.916
TNIE	2	0.83	2.35	50.89	94.83	48.28	2.81	0.919
TNIE	3	0.90	2.26	62.25	94.84	36.85	2.90	0.919
PNDE	1	0.36	2.69	26.87	94.63	72.77	2.68	0.917
PNDE	2	0.52	2.68	36.60	94.62	62.89	2.69	0.918
PNDE	3	0.68	2.69	47.72	94.63	51.60	2.68	0.924
TNDE	1	0.34	2.68	24.10	94.64	75.55	2.68	0.922
TNDE	2	0.48	2.69	32.86	94.61	66.65	2.70	0.916
TNDE	3	0.64	2.71	42.76	94.59	56.60	2.70	0.926
MINT	1	0.47	1.84	64.42	95.63	35.11	2.52	0.919
MINT	2	0.46	1.89	61.91	95.57	37.62	2.54	0.922
MINT	3	0.47	1.85	64.36	95.60	35.17	2.54	0.922

4.3 H3: is power for the mediated interaction near zero?

No: in our run MINT power rises with n and with the population value ah, whereas Chen reported that "All power for MINT is near zero" (printed p. 121) and that it "never reached the .80 level" (printed p. 148). In the contrast cell, the largest MINT in the design at the smallest sample size, chosen after our full results were available as the most direct test of that statement ($n = 500$, $a = .59$, $h = .34$, $ah = .2006$, averaged over b, c' and cutoffs), our power is 0.836; from $n = 1{,}000$ it is at least .97 whenever $ah \geq .13$ (Table 5). At $n = 500$ the other mediated interactions have power 0.011–0.719. Rejection rates do not use population values, so this disagreement cannot come from how estimates were compared with population values; it could still come from another unprinted step (Section 10). Chen gives no number for the cell, so no tolerance or magnitude label can be computed (row E4 of Table 1); the verdict rests on the qualitative conflict.

4.4 H4: does a sample size of 5,000 suffice?

Mostly, so we partly replicated Chen's tentative recommendation that "a minimum sample size of 5000 has accurate Type I error rates and power above .80, regardless of cutoff score" (printed p. 150). Table 6 shows the share of passing results at each n (`statement_checks.csv`). At $n = 5{,}000$, 86.3% of the 1,176 zero-effect results have Type I error inside Bradley's $[.025, .075]$ and 89.2% of the 2,664 nonzero results have power of at least .80. The Type I share is 70.0% and the power share 35.9% at $n = 500$; $n = 5{,}000$ is the smallest sample size at which both shares exceed 85%, and the largest single gain in the power share (61.5% to 89.2%) comes between $n = 2{,}500$ and 5,000. At $n = 5{,}000$ the shares differ little by cutoff (85.7–87.0% and 88.3–89.9%). Not all results pass: product effects whose two paths are both zero stay

Table 5: MINT power (rejection rate) by sample size and population MINT $= ah$, averaged over b, c' and the three cutoffs. Our simulation (summary.json, field `mint_power_by_n_and_ah`); Chen tabulates no MINT power values.

	ah								
n	.0070	.0195	.0210	.0295	.0476	.0585	.0885	.1326	.2006
500	0.011	0.063	0.049	0.079	0.148	0.249	0.275	0.719	0.836
1,000	0.026	0.111	0.152	0.106	0.272	0.558	0.477	0.974	0.982
2,500	0.113	0.215	0.519	0.178	0.560	0.912	0.836	1.000	1.000
5,000	0.328	0.369	0.845	0.302	0.844	0.995	0.981	1.000	1.000
10,000	0.669	0.622	0.987	0.518	0.988	1.000	1.000	1.000	1.000

conservative, with Type I error about .002, below Bradley's range, at every n, and small effects such as $ah = .007$ stay below .80 power even at $n = 10{,}000$ (0.669). Chen calls the recommendation tentative; our check reads it literally over the whole grid. No scalar contrast represents a recommendation over the grid, so its magnitude is not established (row E5).

Table 6: Share of results meeting each criterion, by sample size, over all effects and cutoffs. Our simulation (statement_checks.csv); 1,176 zero-effect and 2,664 nonzero-effect results at each n.

n	Type I error in $[.025, .075]$ (%)	Power $\geq .80$ (%)
500	70.0	35.9
1,000	74.0	53.3
2,500	80.1	61.5
5,000	86.3	89.2
10,000	89.6	91.4

5 Supporting claims, RQ1 and RQ2

5.1 Indirect and mediated-interaction bias (S1)

Not replicated: Chen reported that PNIE, TNIE and MINT fall inside $[-.10, .10]$ in 51.88–87.50% of conditions, with non-bias increasing with the control share (Table 4.8), and all of ours lie inside $[-.05, .05]$ (Table 3). Averaged over cutoffs, her shares outside $[-.10, .10]$ are 27.08 (PNIE), 39.06 (TNIE) and 12.50 (MINT) percentage points (our calculation from her printed values), against 0.00 for each in our run.

zero_estimate_check.py asks what Table 4.8 would show if every estimate were zero, so that |bias| equals the population value over the 256 combinations of a, b, c' and h (zero_estimate_check.csv). At the median cutoff, Chen's PNIE row (75.00% and 50.00%), her MINT row (87.50% and 75.00%, as at all three cutoffs) and her TNIE $[-.05, .05]$ entry (42.19%) equal what that pattern gives; her TNIE $[-.10, .10]$ entry is 60.00% against the pattern's 59.38%. The pattern does not fit every printed entry: PNDE would need a magnitude of $c' + h$ rather than its population value c' to give her 12.50% inside $[-.10, .10]$, and even then not her 6.25–10.00% inside $[-.05, .05]$. This is a consistency observation about the printed table, not a diagnosis of what her code did.

5.2 Type I error and power statements (S2, S3)

Chen's Type I error and power statements partly replicate. `statement_checks.py` tests 46 statements from her Type I error section (printed pp. 99–108) and power section (printed pp. 110–121): 18 on Type I error and 28 on power (Table 7). Seven further sentences in those sections were considered and not tested, because they describe a trend, summarize statements tested elsewhere, or offer an explanation rather than a result; they are listed with their reasons in Table 8. Each tested statement was tested on every cell set it names, for example TNIE1 at $b = .14$, .39 and .59, or MINT with $a = 0$ at each $h > 0$, with rates averaged over the factors it does not name, as in her figures; a statement agrees only if every cell set agrees. Her words are read with thresholds stated in the docstring of `statement_checks.py` (for example, a "near zero" Type I rate is below .025, power "close to zero" is below .10, and power "around .80" is .70–.90); these readings are ours, and a second verdict, which moves every threshold .01 in her favour, shows which verdicts depend on them. Statements 25–29 come from a paragraph that refers to Figure 4.23 (TNIE1 at every h) but describes TNIE at $h = .15$ across cutoffs (Figure 4.24); the script tests their intersection, cutoff 1 at $h = .15$, and the other readings were not tested. Statements 30 and 44 read her remarks that the cutoff 2 and 3 results for TNIE and TNDE are similar to cutoff 1 as no differences across the three cutoffs. Statement 43 tests the TNDE cells with $c' > 0$; the $c' = 0$ lines of her Figure 4.27 are not included. Her topic sentence on printed p. 114, that power varied by cutoff for $a = .39$ and .59 with $b = .14$, introduces statements 32 and 33, which test it. Three further sentences are the sources of tested statements or restate them: TNIE power at the other levels of h does "not differ" (printed p. 112; tested as statement 31), the TNDE results for the other levels of h "do not differ" (printed p. 118; the source of statement 45), and there is "very little power to detect MINT" (printed p. 121; restated and tested as statement 46). Because the population PNDE is c' in her data-generating model, the cells of statements 34 and 35 ($c' = 0$, $h > 0$) are zero-effect cells in our run, so their agreement reflects a Type I rate near .05 read against her "close to zero" (below .10) and "below .80".

Of the 46 statements, 29 agree (31 with the slack): 13 of the 18 Type I statements (14) and 16 of the 28 power statements (17). All six PNIE and all five PNDE power statements agree. Fifteen statements disagree under both readings. Among the Type I statements these are MINT with $a = 0$ (statement 18), where our rate is below Bradley's range at $h = .05$ through $n = 5{,}000$ (at most .019) but .026–.036 at $n = 10{,}000$, below the range at $h = .15$ at $n = 500$ (.014–.020), and about .05 at $h = .34$ and at $h = .15$ from $n = 2{,}500$ (.026–.033 at $n = 1{,}000$), whereas she reports near-zero rates with no variation across h; and three TNIE statements with $a = 0$ at small n (statements 9–11), where our rates and her description fall on different sides of the robustness bounds. Among the power statements they are MINT power (statement 46, the H3 headline); TNIE1 power with $b = 0$, which she reports "close to zero across all levels of a" (printed p. 112) and which in our run is 0.122, 0.586 and 0.655 at $n = 500$ for $a = .14$, .39 and .59 and at least 0.849 from $n = 5{,}000$ (statement 25); TNIE power across levels of h and with $a = .39$ or .59 and $b = .14$ (statements 27, 31 and 33); TNDE1 power (statements 39–41); TNDE1 power across levels of h, which she reports does not differ (statement 45); and her remarks that TNIE and TNDE power at cutoffs 2 and 3 is similar to cutoff 1 (statements 30 and 44), whereas our power differs by cutoff, for example 0.940, 0.850 and 0.564 at cutoffs 1, 2 and 3 for TNIE at $n = 1{,}000$, $a = .39$, $b = 0$, $h = .15$.

As a consistency check, `statement_checks.py` also reads every tested statement, with the strict thresholds, on our cells with h replaced by 0 and the other factors unchanged (the "If $h = 0$" column of Table 7). Read this way, 39 of the 46 statements hold, against 29 as designed: 10 of the 15 statements that disagree under both readings become agreements, none that agrees becomes a disagreement, and statements 10, 11, 30, 41 and 44 still disagree. Among the Type I statements, 9 and 18 would hold. Chen also defines PNDE as "adding c' and h" (printed p. 104), whereas with the mediator intercept zero in her data-generating model the population PNDE is c'. Both observations are consistent with h contributing little to her estimates or population values. They are readings of our own results, not tests of her runs, and they do not establish a cause. Over all 5,880 zero-effect results, our mean Type I error is .043 and 80.0% lie inside

[.025, .075]. Only the text was tested; her Figures 4.17–4.28 were not digitized. Agreement on PNIE and PNDE is consistent with Chen's rejection rates for those effects not being affected by whatever produced her bias and coverage results.

Table 7: The 46 statements tested from Chen's Chapter 4: 18 on Type I error (1–18) and 28 on power (19–46), by printed page (PDF page in parentheses), numbered as in the `number` column of `statement_checks.csv`, which lists the rates by n and cutoff for every cell set. Each statement was tested on every cell set it names, with rates averaged over the factors it does not name; it agrees only if every cell set agrees. "Slack" repeats the verdict with every threshold moved .01 in Chen's favour. "If $h = 0$" reads the statement, with the strict thresholds, on our cells with h replaced by 0 and the other factors unchanged; it is a reading of our own results, not a test of her runs. A trailing digit names the cutoff index, as in Chen: PNIE1 is PNIE at cutoff 1, MINT2 is MINT at cutoff 2. Seven further sentences were considered and not tested (Table 8).

#	Page	Claim	Statement	Cell sets	Verdict	Slack	If $h = 0$
1	99 (113)	S2	PNIE Type I below the robustness criteria, near zero, when a = b = 0	1 of 1	agree	agree	agree
2	99 (113)	S2	PNIE Type I outside the robustness criteria at n = 500 and 1000, within for larger n, when b = 0 and a = .14, at all cutoffs	0 of 1	disagree	agree	disagree
3	99 (113)	S2	PNIE Type I within the robustness criteria at all n when b = 0 and a = .39 or .59	2 of 2	agree	agree	agree
4	99 (113)	S2	PNIE1 Type I outside the robustness criteria at n = 500, approaching .05 (within) as n increases, when a = 0 and b = .14	1 of 1	agree	agree	agree
5	99 (113)	S2	PNIE1 Type I within the robustness criteria when a = 0 and b = .39 or .59	2 of 2	agree	agree	agree
6	99 (113)	S2	PNIE2 and PNIE3 Type I within the robustness criteria when a = 0 and b = .14, .39 or .59	3 of 3	agree	agree	agree
7	102 (116)	S2	TNIE Type I below the robustness criteria, close to zero, when a = b = h = 0	1 of 1	agree	agree	agree
8	102 (116)	S2	TNIE Type I approaches .05 (within at the largest n) when a = 0 and b = .14, at every h	4 of 4	agree	agree	agree
9	102 (116)	S2	TNIE3 Type I outside the robustness criteria at n = 500 when a = 0 and b = .14, at every h	2 of 4	disagree	disagree	agree
10	102 (116)	S2	TNIE3 Type I outside the robustness criteria at n = 1000 when a = 0, b = .14, h = 0 or .05	0 of 2	disagree	disagree	disagree
11	102 (116)	S2	TNIE Type I within the robustness criteria in all other cases with a = 0 (b or h nonzero, apart from the TNIE3 exceptions)	12 of 15	disagree	disagree	disagree
12	104 (118)	S2	PNDE Type I within the robustness criteria and close to .05 when $c' = h = 0$	1 of 1	agree	agree	agree
13	104 (118)	S2	PNDE Type I shows no differences across cutoffs when $c' = h = 0$	1 of 1	agree	agree	agree
14	106 (120)	S2	TNDE Type I within the robustness criteria and close to the targeted .05 when $c' = h = 0$, at every level of a	4 of 4	agree	agree	agree
15	108 (122)	S2	MINT Type I close to zero and below the robustness criteria when a = h = 0	1 of 1	agree	agree	agree
16	108 (122)	S2	MINT Type I below the robustness criteria when a = .14, h = 0: MINT1 and MINT3 at n = 500, 1000, 2500; MINT2 at n = 500, 1000	1 of 1	agree	agree	agree
17	108 (122)	S2	MINT Type I within the robustness criteria when h = 0 and a = .39 or .59	2 of 2	agree	agree	agree
18	108 (122)	S2	MINT Type I below the robustness criteria and close to zero when a = 0, with no variation across h > 0	0 of 3	disagree	disagree	agree

#	Page	Claim	Statement	Cell sets	Verdict	Slack	If $h = 0$
19	110 (124)	S3	PNIE power below .80 at n = 500, 1000, 2500 when a = b = .14, with PNIE1 the lowest of the three cutoffs	1 of 1	agree	agree	agree
20	110 (124)	S3	PNIE power below .80 at n = 500, 1000, 2500 when a = .14 and b = .39 or .59, with no differences across cutoffs	2 of 2	agree	agree	agree
21	110 (124)	S3	PNIE, a = .39, b = .14: below .80 at n = 500 at all cutoffs with PNIE1 lowest and PNIE3 highest; at n = 1000 PNIE2 and PNIE3 at or above .80 while PNIE1 stays below	1 of 1	agree	agree	agree
22	110 (124)	S3	PNIE power around .80 at n = 500 when a = .39 and b = .39 or .59	2 of 2	agree	agree	agree
23	110 (124)	S3	PNIE, a = .59, b = .14: below .80 at n = 500 with PNIE1 lowest and PNIE3 highest; at n = 1000 only PNIE3 above .80; at n = 2500 all three above .80	1 of 1	agree	agree	agree
24	110 (124)	S3	PNIE power above .80 at all n and cutoffs when a = .59 and b = .39 or .59	2 of 2	agree	agree	agree
25	112 (126)	S3	TNIE1 (h = .15) power close to zero when b = 0, at every level of a	0 of 3	disagree	disagree	agree
26	112 (126)	S3	TNIE1 (h = .15) power below .80 for n $\leq$ 2500 and above .80 for n $\geq$ 5000 when a = .14 and b = .14, .39 or .59	3 of 3	agree	agree	agree
27	112 (126)	S3	TNIE1 (h = .15) power below .80 at n = 500 and above .80 at larger n when a = .39 or .59 and b = .14	1 of 2	disagree	disagree	agree
28	112 (126)	S3	TNIE1 (h = .15) power around .80 at n = 500 and above .80 at larger n when a = .39 and b = .39 or .59	2 of 2	agree	agree	agree
29	112 (126)	S3	TNIE1 (h = .15) power above .80 at all n when a = .59 and b = .39 or .59	2 of 2	agree	agree	agree
30	112 (126)	S3	TNIE2 and TNIE3 power show overall patterns similar to TNIE1 (read as: no differences across the three cutoffs; h = .15, every a and b)	7 of 12	disagree	disagree	disagree
31	112 (126)	S3	TNIE1 power shows no clear variation across levels of h (h = .05, .15, .34; rates shown for the cell set averaged over h)	7 of 9	disagree	disagree	agree
32	114 (128)	S3	TNIE (h = .15), a = .39, b = .14: TNIE1 highest and TNIE3 lowest; all below .80 at n = 500; TNIE1 and TNIE2 above .80 at n = 1000	1 of 1	agree	agree	agree
33	114 (128)	S3	TNIE (h = .15), a = .59, b = .14: below .80 at n = 500; at n = 1000 only TNIE1 above .80; all above .80 for n $\geq$ 2500	0 of 1	disagree	disagree	agree
34	116 (130)	S3	PNDE power close to zero when $c' = 0$, at every level of h (h > 0) and cutoff	3 of 3	agree	agree	agree
35	116 (130)	S3	PNDE power below .80 for n $\leq$ 2500 when $c' = 0$, at every level of h (h > 0) and cutoff	3 of 3	agree	agree	agree
36	116 (130)	S3	PNDE power slightly below .80 at n = 500 when $c' = .39$, at every level of h and cutoff	4 of 4	agree	agree	agree
37	116 (130)	S3	PNDE power above .80 when $c' = .59$, at every level of h and cutoff	4 of 4	agree	agree	agree
38	116 (130)	S3	PNDE power shows no differences across the cutoffs	12 of 12	agree	agree	agree
39	118 (132)	S3	TNDE1 power close to zero when $c' = 0$, regardless of a and h (both nonzero)	1 of 9	disagree	disagree	agree

#	Page	Claim	Statement	Cell sets	Verdict	Slack	If $h = 0$
40	118 (132)	S3	TNDE1 power below .80 for n $\leq$ 2500, reaching .80 only above 2500, when $c' = .14$	11 of 16	disagree	disagree	agree
41	118 (132)	S3	TNDE1 power around .80 at n = 500 and above .80 at larger n when $c' = .39$, at all a and h	13 of 16	disagree	disagree	disagree
42	118 (132)	S3	TNDE1 power above .80 at all n when $c' = .59$, at all a and h	16 of 16	agree	agree	agree
43	118 (132)	S3	TNDE power shows no variation by cutoff (h = .15)	11 of 12	disagree	agree	disagree
44	118 (132)	S3	TNDE2 and TNDE3 power show overall patterns similar to TNDE1 (read as: no differences across the three cutoffs; every c', a and h with a nonzero TNDE)	52 of 57	disagree	disagree	disagree
45	118 (132)	S3	TNDE1 power does not differ across levels of h (h = 0, .05, .15, .34; rates shown for the cell set averaged over h)	6 of 12	disagree	disagree	agree
46	121 (135)	S3, H3	All power for MINT is near zero (below .10)	0 of 9	disagree	disagree	agree

Table 8: Sentences in Chen's Type I error and power sections that were considered and not tested, with the reason recorded in `statement_checks.csv` (rows with verdict "excluded").

Page	Sentence	Reason not tested
100 (114)	Overall, differences in Type I error rate occurred as a function of sample size when one path was zero while the other path was a small effect size.	summary of the PNIE statements tested above (rows on b = 0, a = .14 and a = 0, b = .14); names no cells of its own
110 (124)	Power varied as a function of sample size and effect size. The targeted power was .80.	general description; names no cells or outcome
110 (124)	Generally, as sample size increased, power also increased.	general trend; names no cells
112 (126)	Given a = .14 and b = .14, .39, or .59, power increased as sample size increased.	trend only; the threshold claim in the next sentence is tested (TNIE1, a = .14)
116 (130)	Power increased as sample size increased.	general trend for PNDE after the $c' = 0$ sentence; names no threshold
118 (132)	In the case of $c' = .14$, power increased as sample size increased.	trend only; the threshold claim in the next sentence is tested
121 (135)	Low power may have been due to the fact that MINT is an interaction term.	explanation, not a result

5.3 RQ2: the Head Start Impact Study (S4, S5, S6)

RQ2 is not established, because its data are unavailable; this is a blocked comparison, not a nonreplication. Chen analyzed the HSIS restricted-use file (3,721 children after excluding crossovers and no-shows, 2,697 with a wave-1 spelling score) and reported that synthetic-RDD mediation estimates at three cutoffs are typically smaller than the RCT estimates, with standardized bias outside an acceptable range (Tables 4.6–4.7; printed pp. 145–146), and that density, covariate-balance, residual-correlation and functional-form checks show slight violations of independence and functional form (Tables 4.3–4.5; printed p. 146). The restricted-use files require a signed ICPSR Restricted Data Use Agreement that this project does not hold, and the public download of ICPSR study 29462 contains documentation only (`availability.md`). Tables 4.2–4.7 and Figures 4.1–4.11 therefore cannot be refitted (S4, S6).

What can be checked from printed numbers was checked (S5). `table_arithmetic.py` recomputes all 24

differences and standardized biases (SB) of Tables 4.6–4.7 from the printed RCT and RDD estimates (`table_arithmetic.csv`). All 24 differences match as RCT minus RDD, and all 24 SBs match at the printed precision when the divisor is the printed standard error of the RCT estimate (Section 8.2).

5.4 RQ1 and the unreported power analysis (S7, S8)

RQ1 is not established in this replication because it is conceptual: Chen's Table 4.1 (printed p. 60) integrates mediation and RDD assumptions from the literature, and there is no computation to rerun. We note only that the simulation's data-generating model meets the no-confounding and functional-form assumptions by construction; the synthesis was not evaluated as scholarship (S7). Chen's Chapter 3 also announces "An additional power analysis by standard deviation of the outcome" to compare expected and simulated power (printed p. 58), but no method or result for it appears in Chapters 4–5 or the appendices, so it is not established (S8).

6 Comparison of published and replicated numbers

Table 9 gives, for each claim, the published value, ours, the precision, the numerical agreement and the reconstruction scope. Numerical is *match* when our value rounds half up to the printed value, *close* when it misses by at most one unit in the last printed place, and *mismatch* otherwise; *withheld* marks a comparison without a published number or without our number. Scope is *exact* for the same data, estimator and procedure, *limited* for a reconstruction, and *unavailable* when the input was not obtained. Our simulation is *limited* because it uses the same model, estimators and counts but our own engine and random streams, so it can agree with Chen only on summaries. Each row's reason follows the table.

Reasons, row by row. H1: every direct-effect result lies inside $[-.05, .05]$ in our run. H2 (both rows): our coverage is near nominal in every effect-cutoff row and misses fall on both sides at similar rates. H3: Chen prints no number, and our power reaches 0.836 at $n = 500$ for $a = .59$, $h = .34$ where she reports near zero. H4: at $n = 5{,}000$, 86.3% of zero-effect and 89.2% of nonzero results pass, but both-zero products and the smallest effects do not. S1: all our indirect and MINT results lie inside $[-.05, .05]$. S2: the MINT statement for $a = 0$ disagrees because our rate varies with h where she reports none, and in three TNIE statements with $a = 0$ at small n our rates and her description fall on different sides of the robustness bounds. S3: all PNIE and PNDE power statements agree; some TNIE, TNDE and MINT statements do not. Read with $h = 0$, 39 of the 46 statements would hold in our run. S4 and S6: the restricted-use data are unavailable. S5: all differences and SBs reproduce, with the SE divisor. S7: there is nothing computational to replicate. S8: the analysis is announced but not reported.

Every remaining gap is large: each printed row of Tables 3 and 4 differs from ours by at least 12.50 (non-bias) and 11.14 (coverage) percentage points, against a single-cell binomial standard error of about .007 in proportion (0.7 percentage points). None of these is a rounding, typographic or boundary difference.

Small differences, which are not replication failures, are listed separately. (i) The cutoff labels disagree between Chen's code and parts of her prose (Section 8.1). (ii) Her SB divisor in the text differs from the one that reproduces her tables (Section 8.2).

7 Assumptions

Each subsection states a choice we supplied because the dissertation does not state it, and what it affects.

Table 9: Claim-level comparison. Published values are Chen's, or our calculation from her printed tables where marked †; our values come from the files in Appendix A.

Claim	Published	Ours	Precision	Numerical	Scope
H1 direct effects, % with |bias| > .10	85.42†	0.00	2 decimals	mismatch	limited
H2 coverage shortfall from 95%	43.62†	0.04	2 decimals	mismatch	limited
H2 imbalance, above minus below	47.07†	0.33	2 decimals	mismatch	limited
H3 MINT power, $n = 500$, $ah = .20$	near zero (verbal)	0.836	verbal	withheld	limited
H4 minimum n of 5,000	qualitative	Type I 86.3%, power 89.2% pass	not applicable	not applicable	limited
S1 PNIE, TNIE, MINT % with |bias| > .10	27.08, 39.06, 12.50†	0.00 each	2 decimals	mismatch	limited
S2 Type I statements	18 statements	13 of 18 agree (14 with slack)	thresholds	not applicable	limited
S3 power statements	28 statements	16 of 28 agree (17 with slack)	thresholds	not applicable	limited
S4 HSIS RDD versus RCT	typically smaller	not rerun	–	withheld	unavailable
S5 Tables 4.6–4.7 arithmetic	24 entries	24 reproduced	3 decimals	match	exact
S6 HSIS assumption checks	slight violations	not rerun	–	withheld	unavailable
S7 Table 4.1 synthesis	conceptual	not applicable	–	not applicable	not applicable
S8 power by SD of Y	not reported	not rerun	–	withheld	unavailable

7.1 Population values

Chen does not print the population values against which bias and coverage are computed. We derived them from her estimands (equations 3.7–3.11) under her data-generating model, in which the m model has no intercept, so $i_{0M} = 0$: PNIE $= ab$, TNIE $= ab + ah$, PNDE $= c'$, TNDE $= c' + ah$ and MINT $= ah$ (`population()` in `chen_sim.py`). This choice affects bias and coverage but not rejection rates.

7.2 Aggregation over conditions

Chen states that Tables 4.8 and 4.9 are "collapsed across sample and effect size" (printed p. 87) but does not print the aggregation code. We take each table entry as a share or mean over all 1,280 conditions for that effect and cutoff index, with bias bounds inclusive ($|\text{bias}| \leq .10$) and coverage counting an interval whose endpoint equals the population value as covering.

7.3 Zero and nonzero effects

A result counts toward Type I error when its population value is exactly zero and toward power otherwise (`aggregate.py`). This follows Chen's definition (printed p. 58); the choice we supplied is the numerical test for zero.

7.4 Random numbers

Chen's R random streams cannot be reproduced without R, so each condition uses its own NumPy seed, `SeedSequence(20211201, spawn_key=(k,))`. Agreement is therefore judged on summaries, not on individual draws.

7.5 Headline contrasts and the MINT cell

The headline contrasts E1–E3 are averages over the rows of Chen's Tables 4.8 and 4.9, so that one signed number summarizes each headline; their tolerances follow from her two-decimal printing. For H3 we chose the largest MINT in the design at the smallest sample size ($n = 500$, $a = .59$, $h = .34$) as the contrast cell, after the full results were available, because it is the most direct test of "All power for MINT is near zero"; `results/summary.json` and Table 5 give MINT power for every n and ah.

7.6 Literal reading of the sample-size recommendation

Chen calls the minimum sample size of 5,000 tentative. We test it literally, over every zero-effect and nonzero-effect result in the grid at that sample size, which is the strictest reading.

8 Ambiguities that were resolved

8.1 Cutoff labels

The ambiguity is which share of units is treated at each cutoff index. Chen's code (Appendix A, printed pp. 175–176) sets the first cutoff at the 25th sample percentile and assigns treatment to units at or above it, so index 1 has 75% treated; her Chapter 4 text (printed p. 86) agrees, labeling index 1 "25% control—75% treatment". Her Table 4.8 note (printed p. 88) and her Chapter 3 design description (printed p. 55) instead label index 1 as 25% treatment and 75% control. We checked the code against the prose and ran the design as coded; all our tables report by the code's index, with index 3 the 75th percentile and 25% treated (`chen_sim.py`, `table48_ours.csv`). The labels do not affect any verdict, because our bias and coverage show no trend across cutoffs, but readers comparing cutoff-specific statements should use the code's index.

8.2 Standardized-bias divisor

The ambiguity is the divisor of Chen's standardized bias. Equation 3.12 (printed p. 54) divides the RDD-minus-RCT difference by "the standard deviation of the RCT outcome". Recomputing the 24 printed SBs of Tables 4.6–4.7 shows that all 24 match only when the divisor is the printed standard error of the RCT estimate; for example, for path a at cutoff 1, $(1.301 - 2.066)/0.733 = -1.044$, the printed value (`table_arithmetic.csv`). The tables are therefore internally consistent with the SE divisor. Which divisor Chen intended is unknown, so the text and the tables disagree; without the HSIS data the choice cannot be tested further.

8.3 Engine fallback from R to Python

The ambiguity is whether the replication can use Chen's engine. R (`Rscript`) could not be installed on either computer used for the run (`fallback.json`). Chen's analysis is OLS with `lm()` and percentile

bootstrap CIs, with no structural-equation model, so a Python and NumPy re-implementation can express it exactly. The check was the independent validation of Section 3, which passed all three of its tests. The implication is that the replication is distributional: it reproduces Chen's procedure, not her random draws.

9 Limits of the evidence

The evidence has seven limits, each also stated where it applies.

1. R is unavailable here, so the simulation is an independent Python re-implementation with our own seeds; it agrees with Chen on summaries only (H1–H4, S1–S3).
2. Chen's aggregation code and population-value table are not printed, so the source of her bias, coverage and MINT results cannot be identified (H1, H2, H3, S1); our Type I error and power agree with most of her statements (S2, S3).
3. RQ2 (HSIS) cannot be rerun: the restricted-use data are not available to this project (S4, S6); only the printed arithmetic was checked (S5).
4. The announced power analysis by the SD of Y is not reported, so it cannot be assessed (S8).
5. The dissertation labels the cutoffs inconsistently: the code's index 1 is the 25th percentile (75% treated), as is the Chapter 4 description of the cut scores, while Chapter 3 and the Table 4.8 note call index 1 25% treatment; we report by the code (H1, H2).
6. Chen suggests global estimation with an incorrect functional form as a plausible explanation of the bias; in her data-generating model the assignment variable does not enter M or Y, so the fitted linear models are correctly specified (H1).
7. Read on our cells with h replaced by 0, most of Chen's Type I and power statements that disagree with our run would hold, and her text defines PNDE as $c' + h$ where the population value is c'; both are consistent with h contributing little to her estimates or population values, but neither is tested against her runs (S2, S3, H3).

10 What remains unresolved

Four gaps remain open. For each we give the candidate causes, what each check gave, the rank of the remaining causes, and the artifact that would settle the largest.

10.1 Where Chen's bias and coverage figures arise (H1, H2, S1)

Candidates and checks. (a) A coding error in our estimator: ruled out by the scikit-learn and explicit-resampling validation and by the agreement of our estimator with the models of Appendix B. (b) Global estimation with an incorrect functional form, Chen's own explanation: not supported, because her data-generating model makes the fitted models correctly specified (limit 6). (c) Monte Carlo error: ruled out, because the gaps exceed it by orders of magnitude. (d) An engine difference between R and NumPy: not supported, because the estimator is plain OLS and all of her PNIE and PNDE power statements agree with our rates; the TNIE, TNDE and MINT statements that disagree are the subject of Section 10.2. (e) An unprinted step that compared her estimates with population values or aggregated the cells: untested, because that step is not printed; the zero-estimate pattern that reproduces several of her Table 4.8 entries is a consistency observation, not a diagnosis. *Rank.* Candidate (e) ranks first only because it is the one

candidate not contradicted by the evidence; it is untested, not established. Candidates (a)–(d) are ruled out or unsupported. *Stopping statement.* The full design was completed (all 3,840 cells at the printed replication and bootstrap counts, 13.7 core-hours); Chen's own R output and seeds were not available. Her result files or aggregation code, with her per-replication CIs and population values, would show where the printed bias and coverage arise.

10.2 Why many of Chen's Type I and power statements disagree (H3, S2, S3)

Candidates and checks. Rejection rates do not use population values, so the comparison with population values in candidate (e) cannot explain them; an unprinted aggregation of the cells, the other part of (e), could. Fifteen of the 46 statements we tested from her text disagree with our run under both readings of her words. Read on our cells with h replaced by 0, 10 of those 15 would hold and none of the agreeing statements would fail; statements 10, 11, 30, 41 and 44 disagree either way. For MINT Type I error with $a = 0$, ours is near zero, below Bradley's range, at $h = .05$ through $n = 5{,}000$ and at $h = .15$ at $n = 500$, but about .05 at $h = .34$ and at $h = .15$ from $n = 2{,}500$, so it varies with h where hers does not. Her text also defines PNDE as $c' + h$, where the population value is c'. These observations are consistent with smaller h estimates or wider h intervals in her runs than the printed code produces, or with h contributing little to her population values; they are readings of our results and are not tested against her runs. Engine differences are not supported, for the reason given in (d) above. *Rank.* Smaller h estimates or wider h intervals rank first, because they are consistent with 10 of the 15 disagreements; an unprinted aggregation step ranks second; an engine difference is not supported. None is established. *Stopping statement.* The full design was completed; Chen's h estimates and the CIs for one condition with $h > 0$ would show whether they were smaller or wider than ours.

10.3 The HSIS analyses (S4, S6)

Stopping statement. The input is unavailable: the ICPSR 29462 public download V7 contains documentation only, and the restricted-use route needs a signed agreement. All HSIS fits are untested. The settling artifact is the restricted-use HSIS file under an ICPSR agreement.

10.4 The announced power analysis (S8)

Stopping statement. Chapters 3–5 and Appendices A–B were searched for "expected power", "standard deviation" and "power analysis"; no method or result is reported. The settling artifact is Chen's analysis notes. A related, smaller gap is that the Type I and power statements of S2–S3 were checked from the text; digitized Figures 4.17–4.28 (Type I error in Figures 4.17–4.21, power in Figures 4.22–4.28) would allow a cell-by-cell numerical comparison.

11 Code and source availability

The replicated work is Chen (2021), available at `https://digitalcommons.unl.edu/cehsdiss/403/`; the copy read here, `source/chen2021.pdf`, is not in the repository and can be downloaded from that record (SHA-256 `9206b3d824077d36c86b4935ff7f19f64b3fdebde7746943b8a94500fe11e557`). All code and summary results are in the study directory (Appendix A), and the study `README.md` gives the reproduction commands, seeds and outputs. The per-condition files of `simulation/work/` are not committed; `aggregate.py` turns them into `cells.csv` and the summary tables, which are committed. The HSIS data are not available to this project. Table 10 is the dated availability ledger from `availability.md`.

Table 10: Source-data availability for RQ2, checked on 2026-11-08 (UTC).

Place checked	Result
Chen (2021), Chapter 3, Dataset	Uses the HSIS restricted-use dataset from ICPSR under agreement; 3,721 children, 2,697 with a wave-1 spelling score.
ICPSR 29462, HSIS 2002–2006	Downloads after free sign-in, but the public download (next row) contains documentation and codebooks only; the data files are restricted-use and need a signed Restricted Data Use Agreement.
ICPSR 29462 public download V7 (2018-02-08)	58 files, 454 MB unpacked: guides, codebooks and questionnaires for DS0001–DS0030; no data files.
ICPSR 36968, HSIS 2002–2008	Related restricted collection; not checked further.
Chen (2021), appendices	No RQ2 analysis data or Mplus input; Appendices A–B cover only the simulation.

12 Next steps

Three artifacts would settle the open gaps, each with a defined check. First, Chen's result files or aggregation code, including her population values and her h estimates and CIs for one condition with $h > 0$: recompute her Tables 4.8 and 4.9 and her MINT, TNIE and TNDE rejection rates from them and compare them, cell by cell, with `cells.csv`. Second, the restricted-use HSIS file under an ICPSR agreement: refit her Tables 4.2–4.7 with her printed models and compare estimates and SBs, under both SB divisors. Third, digitized Figures 4.17–4.28: compare her plotted Type I error rates and power with `type1.csv` and `power.csv` cell by cell, judging each against two Monte Carlo standard errors with a Holm adjustment.

References

James V. Bradley. Robustness? *British Journal of Mathematical and Statistical Psychology*, 31(2): 144–152, 1978. doi: 10.1111/j.2044-8317.1978.tb00581.x.

Donna Chen. *Statistical Mediation Analysis in Regression Discontinuity Design for Causal Inference*. Doctoral dissertation, University of Nebraska-Lincoln, Lincoln, Nebraska, December 2021. URL https://digitalcommons.unl.edu/cehsdiss/403/. Advisers: James A. Bovaird and Lorey A. Wheeler.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Number 57 in Monographs on Statistics and Applied Probability. Chapman & Hall, New York, 1993.

Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

Jeffrey J. Hoogland and Anne Boomsma. Robustness studies in covariance structure modeling: An overview and a meta-analysis. *Sociological Methods & Research*, 26(3):329–367, 1998. doi: 10.1177/0049124198026003003.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre

Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, 2021. URL https://www.R-project.org/. Chen (2021) used R version 4.0.3.

Linda Valeri and Tyler J. VanderWeele. Mediation analysis allowing for exposure-mediator interactions and causal interpretation: Theoretical assumptions and implementation with SAS and SPSS macros. *Psychological Methods*, 18(2):137–150, 2013. doi: 10.1037/a0031034.

Tyler J. VanderWeele. *Explanation in Causal Inference: Methods for Mediation and Interaction*. Oxford University Press, 2015.

A Artifacts and reproduction

Links point to the `master` branch as navigation aids, not as an immutable reproduction version. Commands run from the study root `formalization/chen-rdd-mediation/` unless a `cd` is shown, as in the "Reproduce" section of the study `README.md`. The simulation needs Python 3 with NumPy; the validation also needs scikit-learn; `build_packet.py`, which builds the evidence file, needs `pdftotext` and the source PDF, which is not in git.

Claim	File	Command
Simulation design and estimator (all H, S1–S3)	`chen_sim.py`	`cd simulation && OPENBLAS_NUM_THREADS=1 python3 chen_sim.py run --out work --workers 2` (defaults: 1,000 replications, 500 resamples; seed per condition 20211201 with spawn key k)
Estimator validation	`validate_estimator.py`	`cd simulation && python3 validate_estimator.py` (seeds 7 and 11; prints `PASS_ESTIMATOR_VALIDATION`)
Engine fallback	`fallback.json`	record; no command
Cell results (19,200 rows)	`cells.csv`	`cd simulation && python3 aggregate.py work ../results`
H1, S1: Table 4.8 analogue	`table48_ours.csv`	same command
H2: Table 4.9 analogue	`table49_ours.csv`	same command
S2, H4: Type I error by n	`type1.csv`	same command
S3, H4: power by n	`power.csv`	same command
H1–H3: overall summaries, MINT power	`summary.json`	same command
S2, S3, H3, H4: statement checks	`statement_checks.py`, `statement_checks.csv`	`cd simulation && python3 statement_checks.py ../results/cells.csv > ../results/statement_checks.csv`
S1: zero-estimate consistency check	`zero_estimate_check.py`, `zero_estimate_check.csv`	`python3 secondary/zero_estimate_check.py > secondary/zero_estimate_check.csv`
S5: Tables 4.6–4.7 arithmetic	`table_arithmetic.py`, `table_arithmetic.csv`	`cd secondary && python3 table_arithmetic.py` (writes the CSV in the working directory)

Claim	File	Command
S4, S6: data availability	availability.md	record; no command
Run record, software versions	run-record.md	record; no command
Plan and part statuses	REPLICATION-PLAN.yaml	record; no command
Evidence file (Chen's claims, our results, locators)	build_packet.py, evidence-packet.json	python3 build_packet.py (needs the source PDF and pdftotext)
Source dissertation (not in git)	Digital Commons record; local copy source/chen2021.pdf	download from the record; SHA-256 9206b3d824077d36c86b4935ff7f19f6 4b3fdebde7746943b8a94500fe11e557
Reproduction recipe, seeds, outputs	README.md	read; section "Reproduce"
Typesetting template and its origin	handoff/SNAPSHOTS.md, approved-preamble.tex	record; no command
This note	mediation_in_regression_discontinuity__a_replication_of_chen_2021.tex, references.bib	cd paper; pdflatex; bibtex; pdflatex; pdflatex (TeX Live 2023)
Directories	simulation/, results/, secondary/, notes/	–

Rebuilding this note from the saved outputs needs only the last command. Rerunning the simulation regenerates simulation/work/ (one JSON file per condition, written atomically, so an interrupted run resumes by skipping finished conditions) and then the results files; it takes about 13.7 core-hours. Re-extracting source data is not possible for RQ2.

B Compliance ledger

Each row is a reporting standard this note follows, whether it was met (*followed*) or not (*departed*, with the reason), and where to check it.

Standard	Status	Evidence or reason
The abstract and introduction open with the verdict, cite the source and state its finding	followed	Abstract; Section 1
The limit of the evidence follows the verdict	followed	Abstract; Section 1
The blocked RQ2 comparison is not called a nonreplication	followed	Abstract; Section 5
No cause of Chen's discrepancies is stated as established; the zero-estimate pattern is a consistency observation	followed	Sections 4.1, 5 and 10
Exact run counts (3,840 of 3,840 cells) in abstract and body	followed	Abstract; Sections 1, 3
Page-1 table with signed effects, magnitude, direction, verdict, tolerances and units	followed	Table 1
Claim-level table with numerical agreement and scope	followed	Table 9

Standard	Status	Evidence or reason
Monte Carlo error and multiplicity considered	followed	Section 4.2: every gap exceeds any two-standard-error band, with or without a Holm adjustment
Assumptions and resolved ambiguities in their own subsections	followed	Sections 7, 8
Each open gap has a stopping statement	followed	Section 10
Same computing engine as the source	departed	R was not available on the computers used; the Python and NumPy re-implementation was validated independently (`fallback.json`, Section 8.3)
Software cited with exact versions	followed	Python 3.13.18 with NumPy 2.5.3; Python 3.10.12 with NumPy 2.2.6; scikit-learn 1.9.1 (`run-record.md`); Chen used R 4.0.3
Citations carry page, table and equation locators	followed	Printed page numbers of Chen (2021) throughout
Every reference checked against its original publication	departed	Bradley, Hoogland and Boomsma, Valeri and VanderWeele, VanderWeele, and R Core Team were checked against Chen's reference list; Efron and Tibshirani, Harris et al. and Pedregosa et al. are not in that list, and their entries use standard bibliographic details that were not checked against the publishers
Citation keys resolve and every reference is cited	followed	The repository's `check_citations.py` found 9 citation keys, 9 references and no errors; this structural check does not test whether a source supports a claim, which was checked by reading the cited passages
Repository files linked on `master` with visible filenames	followed	Appendix A
Reproduction tied to a fixed commit	departed	No commit-pinned link is given; `master` links are navigation aids, and the commands in Appendix A and the study `README.md` define the reproduction
A README that runs every script, with seeds and outputs	followed	Study `README.md`, section “Reproduce”
Origin of the typesetting template recorded	followed	`handoff/SNAPSHOTS.md` names the repository file and revision from which `approved-preamble.tex` was copied
Typesetting template used without additions	departed	The template is loaded unchanged, but this note adds the Latin Modern fonts with T1 encoding, `textcomp` and `microtype` with f-ligatures disabled, so that apostrophes, vertical bars and ligatures copy out of the PDF as the characters typed and no bitmap fonts are embedded; Latin Modern is the installed scalable family the template requires before T1 encoding is used
The source PDF is not presented as a repository file	followed	`source/chen2021.pdf` is not in git; the note links the Digital Commons record and gives the file's SHA-256
No author line and no statement of how the note was written	followed	Title block
Every claim and chapter listed, with the scope of sentence-level checks stated	followed	Appendix C; Section 5: 46 statements tested, seven sentences excluded with reasons (Table 8)

C Inventory of claims and dissertation sections

Every claim recorded in the study's evidence file, `evidence-packet.json` (a machine-readable list of Chen's claims with their source locations, our results and their statuses, built by `build_packet.py`), appears below with its status and where this note treats it; every chapter and appendix of the dissertation follows. Printed page numbers equal PDF page numbers minus 14.

Claim	Chen's claim and locator	Status	Treated in
H1	Direct effects biased, mostly negatively (Table 4.8, p. 88; pp. 86–87, 147)	not replicated	Table 1 E1; Sections 4.1, 10
H2	Percentile bootstrap coverage 24.10–83.99%, truth above the CI; CIs invalid (Table 4.9, p. 124; pp. 123, 150)	not replicated	Table 1 E2–E3; Sections 4.2, 10
H3	MINT power near zero, never .80 (pp. 121, 148)	not replicated	Table 1 E4; Section 4.3
H4	Tentative minimum n of 5,000 (p. 150)	partly replicated	Table 1 E5; Section 4.4
S1	Indirect and MINT non-bias 51.88–87.50% (Table 4.8, p. 88)	not replicated	Section 5
S2	18 Type I error statements (pp. 99–108)	partly replicated (13 of 18)	Section 5, Table 7
S3	28 power statements for all five effects (pp. 110–121)	partly replicated (16 of 28)	Section 5, Table 7
S4	HSIS RDD estimates typically smaller than RCT (Tables 4.6–4.7; pp. 145–146)	blocked (not established)	Section 5
S5	Differences and SBs of Tables 4.6–4.7; equation 3.12	replicated (arithmetic)	Sections 5, 8.2
S6	HSIS assumption checks (Tables 4.3–4.5; p. 146)	blocked (not established)	Section 5
S7	Integrated assumption table (Table 4.1, p. 60)	not established (conceptual)	Section 5
S8	Power analysis by SD of Y (p. 58)	blocked (not reported)	Sections 5, 10

Part of the dissertation	Content	Coverage here
Front matter (title page, abstract, acknowledgements, contents)	Abstract summarizes RQ2 and RQ3; it says Type I error and power for all effects reached the robustness criteria, which differs from the Chapter 4–5 MINT power statements	Citation metadata; the abstract's power statement is discussed in Section 1
Chapter 1, Introduction (p. 1)	RQ1–RQ3, hypotheses, Table 1.1 design	H1–H4, S7; the design run here
Chapter 2, Literature review (p. 9)	Potential-outcomes definitions of PNIE, TNIE, PNDE, TNDE, MINT; RDD assumptions	No analysis of its own; definitions give the population values (Section 7.1)
Chapter 3, Methods (p. 48)	HSIS analysis plan, equations 3.1–3.13, simulation design (Table 3.1)	S5, S8; Sections 3, 8.1, 8.2
Chapter 4, Results (p. 59)	RQ1 Table 4.1; RQ2 Tables 4.2–4.7, Figures 4.1–4.11; RQ3 Tables 4.8–4.9, Figures 4.12–4.38	H1–H3, S1–S7; RQ2 blocked; the bias-section prose (Figures 4.12–4.16, pp. 86–98) and the coverage-section prose (pp. 123–142) are covered by the Table 4.8 and 4.9 comparisons, not tested sentence by sentence; 46 Type I and power statements are tested and seven sentences excluded (Tables 7 and 8)
Chapter 5, Discussion (p. 143)	Conclusions on RQ1–RQ3, implications, limitations	H1–H4, S4, S6; the statement that power was uniform across cutoffs for the other effects (p. 148) is covered by the cutoff comparisons in Table 7, not tested separately; her summary that indirect-effect and MINT Type I error was "within the expected robustness interval when at least one path effect size was not equal to zero" (p. 148) was not tested separately and overlaps statements 9–11 and 18
References (p. 155)	Bibliography	Used to verify cited metadata
Appendix A (pp. 175–178)	R code for data generation	Re-implemented (H1, H2); Section 8.1
Appendix B (pp. 179–189)	R code for estimation and percentile bootstrap	Re-implemented (H1–H3)